

Scheduled Air Service as Coordinated Resource Markets: An Industrial-Engineering Reading of the Bonded Architecture

Alessandro Daliana*

May 2026

Abstract

Scheduled passenger air service is coordination across a set of resource markets: crew-hours and certifications, aircraft-time, fuel volume, gate slots, catering counts, maintenance hours, ground handling. Today these markets are internalized by airlines, with the firm shape arising as the conventional architecture’s response to the high transaction and enforcement costs of coordinating across markets without a firm-shaped container. The bonded settlement primitive of the companion mechanism paper changes those costs: enforcement is bilateral and self-enforcing through asymmetric bonding, and atomic resolution makes a multi-edge process tree resolvable from a single signature. Under the bonded primitive the resource markets become directly accessible to the passenger through bonded commerce.

Two architectural properties carry the contribution: *asymmetric bonding within a single process* (the passenger as **rootBuyer** posts $2P_i$ at each commit to each resource provider, and each resource provider posts $2G$ against the cumulative process value at the moment they commit; the kernel enforces a single **rootBuyer** across every order in the process) and *buyer-initiated atomic resolution* (the passenger settles the entire process at destination as a single action). Together they produce two consequences for scheduled-service coordination. First, the resource markets become directly accessible to the passenger: the firm-shaped airline becomes structurally optional because the cross-market coordination it internalizes is coordinatable through bonded commerce. Second, when operational disruption occurs the seller cohort faces direct economic incentive — under locked bonds across every edge — to determine and pay compensation directly to the buyer, before any external mechanism (specialized contracts, dispute forums, insurance, regulators) engages. The contribution is not a better airline; it is a different way of organizing scheduled air service in which the firm-shaped container is structurally unnecessary because the resource markets it internalizes are directly coordinatable through bonded commerce. Cascading-delay reduction follows as a consequence of bond posture and atomic resolution; it is a property the architecture produces, not the property the paper is for.

Keywords: scheduled-service coordination, supply-chain coordination, process modeling, resource markets, weakest-link coordination, industrial engineering

1 Introduction

Scheduled passenger air service is coordination across a set of resource markets. The journey a passenger purchases is the joint output of a flight crew (with required certifications and duty-time bands), an aircraft and its operator, a quantity of jet fuel, slot allocations at origin and destination gates, per-passenger catering, a share of the day’s maintenance allocation for the operating airframe, and ground handling for boarding and baggage. Each is a distinct resource the journey draws on; each is supplied by a distinct set of operational parties; each has natural units of measure already in industry use — crew-hours, type ratings, fuel volume in gallons or

*Corresponding author. adaliana@figaro.org

kilograms, gate-slot time bands, meal counts, maintenance hours, baggage counts. Air-traffic-control services are a further input the flight consumes, but they are public-authority outputs not market-traded; we list them only to bound the resource-market inventory. Each resource market currently operates under its own structural constraints (some auction-based, some bilateral, some near-monopolistic at congested airports); the inventory below makes no claim that those structures are uniform.

Today these resources are coordinated through airlines. The airline contracts with crew agencies, fuel distributors, airport authorities, catering firms, maintenance providers, and ground-handling sub-contractors; sells tickets to passengers; and absorbs the residual coordination cost. The firm shape is the conventional architecture’s response to a coordination problem whose transaction and enforcement costs make firm-internalization cheaper than market access for end customers [Williamson, 1985]. The operations-research literature on airline scheduling and operations [Barnhart and Cohn, 2004, Ball et al., 2007, Lan et al., 2006] largely takes this shape as given; analyses focus on optimization within the firm-internal portfolio rather than on whether the firm-internal organization is itself the right architecture.

The bonded settlement primitive of the companion mechanism paper changes the calculus. Enforcement of bilateral commitments is self-enforcing through asymmetric bonding; multi-party coordination is resolvable from a single signature through atomic resolution; the transaction-cost component that made firm-internalization cheaper than direct market access becomes a fixed capital lockup at the bonded edge. Under that change, the resource markets that today’s airlines internalize become directly accessible to the passenger through bonded commerce. The journey is composed as a process tree of bonded commitments across those markets; each resource provider bonds against the passenger’s process; the passenger settles the entire tree at destination through one atomic `resolveProcess` call.

Paper organization. Section 2 summarizes the settlement primitive. Section 3 develops the architectural properties applied to scheduled-service coordination: per-edge asymmetric bonding, buyer-initiated atomic resolution, the two consequences for the assembly, the kernel / composability-stack distinction, and where specialized mechanism contracts can be developed for specific resource markets. Section 4 treats disruption resolution: the seller-cohort compensation mechanism that arises from buyer-dominance under bond pressure (the primary mechanism), illustrated through a worked bond-posture example and through the crew-allocation case where today’s airline-internal auction produces a documented misalignment that the bonded edge eliminates, with the fallback mechanisms (specialized contracts, coordination tokens, off-chain dispute forums, insurance, regulators) treated downstream. Section 5 treats the schema requirements: one new schema, with the rest of the assembly satisfied by existing infrastructure. Section 6 treats what the architecture preserves (safety regulation, crew-labor protections, insurance, competition law), what it changes (the cost-flow path and the firm-internalization), and the non-cost reasons firms might persist under the architecture nonetheless. Section 7 states the scope conditions. Section 8 concludes.

2 The Settlement Primitive

The bonded primitive is developed formally in the companion mechanism paper. We summarize the features the present argument uses.

Asymmetric bonding (Mechanism 1). For a transaction with payment $P > 0$ and cumulative upstream value $G \geq P$, the buyer locks $2P$ and the seller locks $2G$. Cooperation is the unique profile surviving iterated elimination of weakly dominated strategies. The mechanism scales to N -party process trees through progressive collateralization: each seller bonds against

the cumulative value of all upstream commitments, producing a mesh of independently secured bilateral edges, each of which carries its own equilibrium.

Buyer dominance with atomic resolution (Mechanism 2). Only the root buyer can trigger resolution, and resolution settles all active orders in the process simultaneously or not at all. The all-or-nothing rule induces a weakest-link subgame among sellers: each seller’s payout depends on universal cooperation across the tree. The architecture produces cooperation pressure of magnitude $P_i + 2G_i$ on every other seller from each seller’s perspective, without explicit communication, governance, or managerial layer.

What this implies for scheduled-service coordination. The bonded primitive’s two mechanisms are exactly the two missing properties in the contemporary scheduled-service architecture. Mechanism 1 supplies a bilateral commitment whose enforcement does not depend on either party’s home jurisdiction’s contract-law regime; this is what makes it possible for the passenger’s commitment to a counterparty and that counterparty’s commitment to its sub-suppliers to be denominated in the same coordinative unit. Mechanism 2 supplies the atomic resolution that propagates a single commit-or-don’t decision through the entire process tree; this is what makes operational damages flow proportionally to the unit that caused them rather than being absorbed at whichever contract has the loosest cap. Neither mechanism is novel to the scheduled-service domain; both are general-purpose settlement properties whose application here is the present paper’s contribution.

3 The Bonded Architecture Applied to Scheduled-Service

The substantive contribution of the bonded primitive to scheduled-service coordination operates at the edge of the bonded process tree, not at any particular position within it. Two architectural properties supply the contribution; both are independent of how the operational arrangement is organizationally drawn.

Asymmetric bonding within a single process. The kernel enforces a strong structural rule: every order in a process has the same buyer (the `rootBuyer`). The passenger commits directly to each resource provider; there is no intermediating coordinator between the passenger and any seller. The first commit creates the process under the passenger’s address; every subsequent commit in the same process extends the cumulative-value accumulator G by the new payment. At each commit the passenger posts $2P_i$ for that specific payment, and the seller posts $2G_{\text{at commit time}}$ — the full cumulative process value at the moment they commit. Bond posture is asymmetric: the passenger’s per-commit bond scales with the immediate payment, while each seller’s bond scales with the cumulative value of every upstream commit in the process. Sellers committing later post correspondingly larger bonds; the last seller bonds against the entire ticket value.

Buyer-initiated atomic resolution. The passenger calls `resolveProcess`, and the entire process settles simultaneously. Every active order in the process is resolved together or none is. The passenger’s signature is the only signature required for settlement, and atomicity guarantees that no subset of orders settles without the rest.

Two consequences for the scheduled-service assembly. The two properties together produce two consequences.

First, *the resource markets named in Section 1 become directly accessible to the passenger through bonded commerce*. Each resource provider — crew operator, aircraft operator, fuel supplier, gate operator, caterer, maintenance provider, ground-handling firm — can be a seller

bonded against the passenger’s process, with payout depending on the passenger’s atomic resolution. Each provider’s interests align with the passenger’s at the bonded commit rather than through the firm-mediated incentive structure conventional airlines impose. The misalignments today’s firm-shaped container produces — between the airline’s cost-minimization objective, the resource provider’s preferences, and the passenger’s outcome — dissolve at the bond. We treat one such misalignment in detail in Section 4: the crew secondary market.

Second, *damage flow tracks bond posture across the assembly’s commits rather than the loosest contractual liability cap*. Each seller carries asymmetric bond exposure proportional to the cumulative process value at their commit; the seller-side total grows super-linearly with the number of commits while the passenger’s posture stays at $2P_{\text{ticket}}$. The atomic-resolution rule operates on this set of bonds. When something goes wrong, every seller’s bond is locked across every commit — which is the structural pressure behind the disruption-resolution mechanism we develop in Section 4.

What the kernel constrains; what the composability stack carries. The kernel reads only per-order facts: buyer (which must be the `rootBuyer` in every order in the process), seller, payment, cumulative-value snapshot at commit, and the atomic-resolution rule covering the process. The kernel-level shape of a single process is therefore tightly constrained: one buyer, one or more sellers, all under one cumulative accumulator, all resolving atomically. Deeper compositions across multiple processes (a seller in process *A* becoming the `rootBuyer` of process *B* to coordinate their own sub-suppliers) are admissible at the kernel layer but resolve process-by-process, not across.

Above the kernel, the protocol-and-runtime composability stack attaches schemas to specific orders (the `scheduled-departure-v1` clause binds to whichever order the parties want to subject to schedule-tolerance enforcement; `handoff-v1` attestations attach to operational-handoff orders), mechanism contracts operate on specific order patterns, assembly specifications declare which schemas attach where and which view definitions render which surfaces, and UIs render the process per the assembly’s declarations. An assembly’s structure, once specified, is a fixed verifiable artifact bound to its full composability stack. The bonded-architecture argument of this paper applies to whichever resource markets the designer composes under one process, with the kernel-enforced structural rule holding throughout.

Specialized mechanism contracts per resource market. Each resource market admits its own market mechanism. The existing `DutchAuction` contract supplies one such mechanism for cases where descending-price coordination is appropriate (fuel procurement is a candidate where the supplier market admits competition). Bilateral commerce-v1 commits cover the cases where the parties contract directly. Future mechanism contracts can be developed under the protocol-extension discipline of the companion runtime-composition paper for resources whose market mechanics deserve their own primitive: a slot-allocation mechanism for gate-time bidding, a crew-dispatch mechanism that respects type-rating and duty-time constraints, a fuel-pool mechanism for shared ramp-side procurement among co-located operators. The mechanism-contract space for scheduled air service is underdeveloped today because the firm-shaped container has absorbed the design choice; under direct market access it becomes a first-class design surface.

4 Disruption Resolution Under Buyer Dominance

We turn to the operational question: when something goes wrong — a delay, a slip, a service failure — how is the disruption resolved under the bonded architecture? The substantive answer is that Mechanism 2 of the kernel (buyer dominance with atomic resolution) creates structural pressure for the seller cohort to determine and pay compensation themselves, directly to the

buyer, before any external mechanism engages. This is the primary damage-management mechanism in the architecture. We treat it first, place a bond-posture worked example, develop the crew-allocation case as a concrete illustration, note the IROps-scale extension, locate the result in the companion mechanism paper’s weakest-link result, and then position the fallback mechanisms in their proper register downstream.

4.1 The primary mechanism: seller-cohort compensation under bond pressure

Bond posture per commit. Every order in a process shares the same `rootBuyer`; the passenger as that `rootBuyer` posts $2P_i$ at each commit to each resource provider, and each resource provider posts $2G_{\text{at commit time}}$ where G is the cumulative process value at the moment of that commit. Sellers committing later in the sequence post correspondingly larger bonds. Aggregated across an N -seller process, the bonded exposure of the seller cohort scales super-linearly with the number of resource markets the assembly draws on.

A bond-posture worked example. Consider a stylized domestic short-haul passenger journey totalling \$250. There is no airline-shaped coordinator: the kernel enforces `buyer = rootBuyer` in every order in a process, so the passenger commits directly to each resource provider. The seven commits below are all part of one process under one `rootBuyer` (the passenger). The kernel maintains a single process-wide cumulative-value accumulator G that grows monotonically as each commit lands ($G \leftarrow G + P_i$). At every commit the buyer posts $2P_i$ and the seller posts $2G_{\text{at commit time}}$. The asymmetry shows up across the sequence: the passenger’s buyer-side bond at each commit scales only with the immediate payment to that seller, while each seller’s bond scales with the full cumulative value of every upstream commit in the process. Sellers committing later post correspondingly larger bonds; the last seller bonds against the entire ticket value. This is progressive collateralization in operation. Table 1 traces the sequence.

Commit (in sequence)	P_i	G after commit	Passenger bond $2P_i$	Seller bond $2G$
Root: passenger $\rightarrow$ aircraft operator	\$65	\$65	\$130	\$130
Sub: passenger $\rightarrow$ fuel supplier	\$45	\$110	\$90	\$220
Sub: passenger $\rightarrow$ crew operator	\$40	\$150	\$80	\$300
Sub: passenger $\rightarrow$ maintenance provider	\$15	\$165	\$30	\$330
Sub: passenger $\rightarrow$ catering	\$5	\$170	\$10	\$340
Sub: passenger $\rightarrow$ ground handling	\$15	\$185	\$30	\$370
Sub: passenger $\rightarrow$ origin airport authority	\$20	\$205	\$40	\$410
Sub: passenger $\rightarrow$ destination airport authority	\$20	\$225	\$40	\$450
Sub: passenger $\rightarrow$ TSA (security screening)	\$6	\$231	\$12	\$462
Sub: passenger $\rightarrow$ federal aviation system	\$19	\$250	\$38	\$500
Total bond locked across the process	—	—	\$500	\$3,512

Table 1: Stylized per-commit bond posture for a domestic short-haul passenger journey under one process. The passenger commits directly to each seller; there is no intermediating coordinator (the kernel enforces `buyer = rootBuyer` in every order). G is the process-wide cumulative-value accumulator, monotonically growing with each commit. The passenger’s bond at each commit is proportional to the immediate payment; each seller’s bond is proportional to the cumulative process value at the moment that seller commits. The last seller to commit therefore posts the largest seller bond, $2G_{\text{final}} = 2 \times \$250 = \$500$. Per-seller payments are illustrative; absolute magnitudes vary with route, aircraft type, and operating context.

Two features of the table carry the structural argument.

The asymmetry between buyer and seller bonds. The passenger’s aggregate locked bond is \$500, which equals $2 \times P_{\text{ticket}}$ — the standard buyer posture. The seller cohort’s aggregate locked

bond is \$2,630, more than five times the passenger’s posture, because each seller bonds against the cumulative G that has grown across the upstream commits. This is not an accident of the example; it is the asymmetric-bonding architecture in operation. As the assembly adds more resource markets, the seller-side total grows super-linearly with the number of commits while the passenger’s side stays at $2P_{\text{ticket}}$.

The magnitude of the seller cohort’s locked bond. The seller cohort’s aggregate locked bond (\$2,630) is more than ten times the ticket payment. A non-resolving passenger holds that full pool locked across every commit in the process. The pressure on the cohort to compensate the passenger and secure resolution is structural in proportion that exceeds the ticket by an order of magnitude.

The order in which sub-orders commit reorders which specific seller posts the largest bond, but does not change the aggregate seller cohort total: it depends only on the set of payments, not the sequence. Whichever seller commits last bonds against the full $G_{\text{final}} = P_{\text{ticket}}$.

Withholding resolution as the buyer’s leverage. The passenger holds unilateral authority to call `resolveProcess`. As long as the passenger does not call, every seller’s bond in the tree remains locked. Each seller faces forfeit of their full $2G_i$ on continued non-resolution. The buyer’s atomic-resolution authority is therefore not a remedy in the conventional sense (it does not pay the buyer anything beyond bond recovery); it is leverage. The buyer can hold up the entire tree’s bond pool by simply not signing.

Sellers determine and pay compensation themselves. When operational disruption occurs and the buyer is left worse off than the agreement contemplated, the seller cohort faces direct economic incentive to make the buyer whole enough that the buyer chooses to resolve. The mechanism unfolds operationally as follows.

Sellers approach the buyer with compensation offers. Compensation takes whatever form the buyer accepts: cash transfer, future service credit, accommodation or meal vouchers, refund of the ticket payment, alternative routing, partial or full refund combined with services. The currency of compensation is at the parties’ discretion — the kernel does not constrain it.

Sellers negotiate among themselves about which seller bears what fraction of the compensation. The unit whose performance materially failed bears more. Operational parties around the failure who could not have prevented it (the catering provider who had no role in a fuel slip; the gate operator whose service was unaffected) contribute less or not at all. The negotiation is real but routine: each seller knows their own bond exposure, their counterfactual payout under cooperative resolution, and their willingness-to-pay against losing the bond entirely.

The buyer accepts the compensation that satisfies them and calls `resolveProcess`. Bonds release across the tree; every seller recovers $2G_i + P_i$ less their share of the compensation; the buyer recovers their bond plus the compensation already paid. The settlement is bilateral and direct.

This happens fast. The negotiation occurs at the operational boundary (the airport gate, the destination airport’s customer service desk, in some cases through the runtime UI’s disruption-coordination surface), not in a court filing six months later. It happens before insurance claims, before dispute-resolution forums, before regulatory engagement. Most disruption resolves through this path because that is where the bond pressure points.

The crew-allocation case. The mechanism is concrete in the crew market, where today’s firm-mediated arrangement produces a documented misalignment. Airlines run bidding systems through which crew select trips by seniority and preference. The bid systems are largely artifacts of pilot-union and flight-attendant collective bargaining: senior crew preserve seniority-allocation order through bargained contracts, and the airline’s discretion in the primary auction is bounded by collective-agreement protections. Crew frequently view the resulting allocations

as imperfectly aligned with their preferences and run secondary markets to trade trips with each other after the primary auction settles. The secondary market is operationally inefficient (it duplicates allocation work) and structurally awkward (it requires the airline to permit and process post-allocation swaps). Its existence is evidence that the primary auction does not align crew interests with the operational outcome that matters to passengers (qualified, well-rested, willing crew on the flight).

Under the bonded architecture, the crew-allocation can operate as a direct market between the passenger and the crew, without an airline-internal primary auction or a post-allocation secondary market at the passenger-coordination layer. A crew member bonds against the passenger's process at the crew-edge; the crew member's payout depends on the passenger's atomic resolution at destination; the crew member's interests align with the passenger's directly through the bonded edge. The passenger-coordination auction and the secondary market both become structurally unnecessary — the bonded edge is the passenger-side market, and the allocation discipline is whichever mechanism contract the assembly designer composes for the crew-market. The collective-bargaining institutions over wage rates, duty-time protections, and seniority among crew remain; they are labor-market institutions the bonded primitive does not touch.

Under disruption — say a crew member cannot fly — the crew-edge bond is locked alongside every other edge. The crew operator and crew member negotiate compensation directly with the passenger (find a substitute crew member at speed, accommodate the passenger on the next flight, refund the leg, or whatever the passenger accepts). Resolution follows compensation. The conventional airline's role as intermediary in this negotiation disappears.

IROps-scale coordination. The single-flight example understates the operational complexity of large-scale irregular operations (IROps). A hub disruption produces simultaneous bonded processes across hundreds of flights, with thousands of bonded edges across the resource-market mesh. The compensation negotiation at IROps scale is operationally heavier than the per-flight case: each affected passenger holds a separate bonded process with its own seller cohort, and many resource providers (crew operators, aircraft operators, ground-handling firms) appear in many trees at once. We do not claim the bonded architecture eliminates this coordination cost. We claim it routes the cost in a structurally different direction. Under the conventional regime, the airline absorbs the IROps coordination internally; sub-suppliers see only their per-incident penalties; passengers see liability-cap absorption. Under the bonded architecture, each affected passenger faces a (busy, but bilaterally coordinated) seller cohort negotiating their compensation directly. Operational tooling for the resource provider — whose bond is locked across many simultaneous trees and who needs to coordinate compensation responses at scale — is itself one of the assembly-design questions we identify as warranting empirical work.

Connection to the companion mechanism paper. The mechanism is the operational form of an analytical result developed in the companion mechanism paper. Under buyer dominance with atomic resolution, each seller bears direct economic interest of magnitude $P_i + 2G_i$ in every other seller's performance, because each seller's payout depends on universal cooperation across the tree (the mechanism paper formalizes this as a weakest-link subgame proposition: cooperation is the unique equilibrium surviving iterated elimination of weakly dominated strategies, induced by atomic resolution operating on the bonded mesh). The proposition's analytical content is the formal characterization of the pressure; the operational content is sellers paying compensation to buyers directly to secure resolution. The same structural shape produces the joint-liability dynamic in Grameen-style microfinance, in which group members chip in to cover a struggling member's installment before the lender forecloses, because each member's continued credit access is exposed to every other member's repayment. The bonded architecture's seller-cohort-compensates-buyer dynamic is the same structural mechanism scaled with chain depth

and operating without the social-substrate prerequisites Grameen requires.

Why this is structural, not procedural. The mechanism is not a courtesy that sellers may or may not extend. It is a direct consequence of the bond posture under atomic resolution. A seller who refuses to participate in compensation negotiation watches their full bond forfeit alongside every other seller's. The compensation mechanism emerges from the architecture because the architecture has made non-cooperation strictly worse than reasonable compensation for every seller in the tree.

4.2 Fallback mechanisms

The primary mechanism resolves most disruption directly. Fallback mechanisms engage when the primary does not, or when the disruption involves a class of issue the primary cannot address. We list them in the order of operational engagement.

Specialized mechanism contracts (protocol layer). Where the assembly composes a specialized mechanism contract — a delay-compensation contract that disburses pre-agreed amounts under specified conditions, a slot-reallocation mechanism, a fuel-pool mechanism — the contract handles its specific coordination automatically. The specialized contract complements the primary mechanism by making specific compensation patterns automatic rather than negotiated.

Coordination tokens (runtime layer). Where the assembly composes around a coordination token, the substrate's operational state registers as token value. Token holders bear operational quality through value impact rather than through direct compensation. Token-value reflection is a diffuse signal operating in parallel with the per-edge compensation mechanism, not a substitute for it.

Off-chain dispute resolution forums. The off-chain agreement named in the bonded commitment binds a dispute forum (state court, international arbitration center, ODR platform, Schelling-point juror system). The forum engages when the seller-cohort compensation does not resolve the dispute — typically because the dispute is substantive in a way bond pressure cannot address (duress, mistake, frustration, illegality, public-policy concerns). The bonded record is the evidentiary input the forum operates on, with the Class A / Class B taxonomy the companion evidence-and-law paper develops: Class A kernel byproducts (commit and resolution events) supply the indisputable settlement record; Class B schema-validated attestations (`handoff-v1` timestamps, `delivery-lifecycle-v1` stage events) supply the substantive performance evidence.

Insurance compositions. A passenger or operational party who wants insurance against risks the bond pool cannot absorb (catastrophic events, multi-day weather disruption, force-majeure events) composes a parallel bonded process with an insurer. The insurer's bond underwrites the risk; insurance settlement is its own bonded event. Insurance composes with the bonded primitive without modifying the kernel.

Regulators. Aviation safety regulators, consumer protection authorities, antitrust enforcement, and labor protection regimes operate alongside the bonded architecture and engage in the matters their authority covers. The architecture does not displace them.

The substantive contribution of the bonded architecture to damage management is that the primary mechanism — seller cohort negotiating and paying compensation directly to the buyer, before external mechanisms engage — is structural, immediate, and operates without any of the fallbacks being needed in most cases. The conventional regime's cascading-delay pathology — damages absorbed by whichever contract has the loosest liability cap, with limited unit-level operational signal [Lan et al., 2006] — reduces under the bonded architecture as a consequence of bond posture and atomic resolution.

5 Schemas

The assembly draws almost entirely on existing schema infrastructure. One new schema is required: a binding from a ticket commitment to a specific scheduled departure / arrival pair, so the resolution path can verify whether the seller delivered against schedule.

Existing schemas. `figaro-commerce-v1` carries the currency, payment, and itemized line-items of every commitment in the assembly — the passenger ticket and every sub-procurement commitment. `figaro-delivery-lifecycle-v1` carries staged progression as a sequence of attestations; the schema’s five-slot structure can be re-used by reinterpretation (boarding initiated, cabin ready, pushback, takeoff, arrival, or whatever projection the assembly designer adopts), without authoring a new schema. `figaro-handoff-v1` carries the operational handoffs between operational parties. `figaro-jurisdiction-v1` carries the off-chain forum-selection clause for international flights, where the parties want to specify a particular dispute-resolution forum. `figaro-geo-v1` carries the origin and destination geohashes where geographic-anchoring is operationally relevant. The GHG family of schemas is available for assemblies that wish to compose carbon-emission attestations into the ticket commitment.

The new schema: `scheduled-departure-v1`. The schema binds a `commerce-v1` ticket commitment to a specific scheduled departure / arrival pair, so the resolution path can verify whether the seller delivered against schedule. The fields of the Layer A spec are:

- `flightDesignator` (string, IATA format, e.g. “NK1234”)
- `scheduledDeparture` (ISO 8601 datetime, with explicit timezone)
- `scheduledArrival` (ISO 8601 datetime, same form)
- `originAirport` (string, ICAO 4-character, e.g. “KORD”)
- `destinationAirport` (string, ICAO 4-character)
- `aircraftType` (string, optional, IATA aircraft code; omitted if the parties reserve substitution rights at the runtime tier)
- `toleranceSeconds` (integer, the agreed slip below which no penalty applies; e.g. 900 for a 15-minute window)

The schema enforces one structural constraint at the validator contract: `scheduledArrival` must be strictly after `scheduledDeparture`. We deliberately do not impose a maximum on `toleranceSeconds`. Scheduled-service operations vary widely in operationally meaningful tolerance windows, and hard-coding a maximum at the validator contract would foreclose legitimate use cases under the first-write-wins binding discipline that makes the `schemaId` permanent. The contracting parties choose the tolerance their operation supports; if a domain-specific maximum is later wanted, it can be enforced at a domain-specific schema (e.g. `figaro-passenger-departure-v1` as a tighter sister of the present schema) without mutating the present binding. The schema is the seed of a *scheduled-service* family; later extensions could cover scheduled ground transport, scheduled deliveries with tight windows, and other scheduled-binding use cases without reauthoring the family.

Schemas the architecture deliberately does not include. Three classes of schema are conspicuously absent. First, there is no force-majeure schema that allows a counterparty to avoid bond loss in weather events. Force-majeure handling is a *composition* concern, not a kernel concern: a party who wants to insure against weather-driven delay composes a parallel insurance process; the kernel neither knows about weather nor adjudicates whether a given event qualifies. The companion mechanism paper’s no-escape-hatches analysis applies here directly. Second, there is no schema that governs which delays count toward bond loss; the **scheduled-departure-v1** tolerance is the kernel’s only such criterion, and any further refinement is for the off-chain forum that adjudicates disputes the parties bring to it. Third, there is no schema that grants any party a unilateral reschedule right; rescheduling is a new commitment with a new schemaId or a new instance, not a mutation of the existing binding. Append-only identity is the discipline.

6 Comparison with the Conventional Apparatus

The architectural change should be located precisely against the existing apparatus to clarify what does and does not change.

What changes. The architecture changes the cost-flow path. Under the contract-of-carriage regime, passenger schedule loss is bounded by per-incident penalty caps and the airline’s internal willingness to issue voucher compensation; the supplier’s liability for sub-service slip is bounded by per-incident contractual penalties; and the residual loss is absorbed by whichever party’s contract has the loosest cap. Under the bonded architecture, schedule loss is allocated proportionally to the party whose bond is exposed at the relevant process layer: sub-supplier bond for sub-service slip, root-counterparty bond for arrival-tolerance violation, passenger bond returned in full on satisfactory delivery. The cost flow is direct, proportional, and tracks the operational layer at which performance failed.

What does not change. Safety regulation is unchanged: the FAA in the US, EASA in the EU, and analogous national civil-aviation authorities continue to certify aircraft, pilot licensing, maintenance procedures, and operational standards. Crew-labor protections are unchanged: pilot-union and flight-attendant collective bargaining and flight-time / duty-time limits under FAR Part 117 in the US (with analogous regimes elsewhere) apply to the crew-supplier subprocess under the bonded architecture exactly as they apply under the conventional regime. Insurance and reinsurance are unchanged in their underlying function; the bonded architecture changes the risk profile aviation insurance markets price (a smaller residual after bond allocation; concentrated risk at whichever edges carry the largest cumulative-value bond posture; a new class of bond-default risk at every operational party in the tree) but does not displace the insurance function. Antitrust and competition law are unchanged: operational-party markets remain subject to anti-collusion enforcement, slot-allocation regimes at congested airports remain governed by the relevant authorities, and code-share arrangements remain subject to the applicable competition regimes.

Non-kernel roles firms might still play. The bonded architecture removes the transaction-cost rationale for firm internalization of scheduled-service coordination. It does not remove other roles firms might play around the bonded process. Three deserve mention. First, FAA Part 121 (and analogous regimes in other jurisdictions) imposes a single-certificate-holder requirement on commercial scheduled passenger operations: the operating airline holds the air carrier certificate, and many regulatory and safety functions attach to that single legal entity. Under the bonded architecture the certificate-holder is one of the resource-market sellers (typically the aircraft-operator role); the certificate function is preserved, but the certificate-holder is no

longer a cross-market coordinator. Second, knowledge accumulation across operational cycles — learning about reliability of specific routings, weather patterns, sub-supplier reliability — has firm-level returns to scale that direct market access does not naturally aggregate; firms retaining this knowledge are likely to out-perform unaffiliated participants on quality, even within the star-shaped bonded architecture. Third, brand identity functions as a quality signal in markets with imperfect ex ante observability of service quality. Brands cannot, however, become kernel-level coordinators in the bonded architecture — the kernel forbids any party other than the passenger from being the buyer of resource-market commits. A brand under the bonded architecture is necessarily off-chain: a discovery surface, a quality-signal aggregator, a multi-process curator that recommends specific resource-provider compositions to passengers, or a service-level guarantor that composes an insurance process in parallel with the passenger’s bonded process. None of these roles re-introduce the firm-shaped cross-market coordinator the architecture is designed to make structurally optional.

What this is not. The architecture is not a better airline. It is a re-architecture of the cost-flow path in scheduled-service coordination. Whether any specific operational party operates well or poorly within the architecture is an operational-management question the architecture is silent on.

The architecture is also not a passenger-protection regime. It is a coordination protocol that has the side effect of routing schedule-loss damages to the passenger more proportionally than the contract-of-carriage regime does. Passenger-protection regimes (the EU’s Regulation 261/2004, the US’s Department of Transportation refund and compensation rules, and analogous regimes elsewhere) continue to operate in their own register and apply to the regulated entity in the relevant jurisdiction. The bonded architecture’s passenger-side benefit is structural rather than regulatory.

7 Scope

The argument bounds itself by several scope conditions.

Asset specificity. The architecture works best where sub-suppliers compete in markets with multiple available alternatives. Where a sub-supplier is a near-monopolist (as airport authorities often are at hub airports for gate operations, or as ATC services are by regulatory construction), the buyer-side leverage at sub-procurement is constrained, and the bonded architecture devolves toward the conventional single-supplier regime. The companion institutional-economics paper’s discussion of asset specificity applies here: the architecture is most effective in the low-specificity region of the resource portfolio and least effective at the high-specificity end.

Bond-currency access. The architecture presumes that all parties can post collateral in a denomination the kernel accepts. For some sub-suppliers (e.g. small catering contractors operating in jurisdictions with limited crypto on-ramps) this may be the binding operational constraint. The companion statelessness paper’s bond-currency discussion applies in attenuated form: the constraint is real but solvable through standard treasury and on-ramp infrastructure for parties with the resources of typical airline sub-suppliers.

Aggregate liquidity at the operational tier. A high-volume operational party operating thousands of flights per day under the bonded architecture would face aggregate bond-custody requirements that scale with daily flight volume. This is an operational treasury question: the party maintains a working bond pool sized to its operational fleet, with bonds released and rebound as processes resolve. The cumulative custody at any moment is bounded by the active-process count multiplied by per-process bond requirements; the architecture is liquidity-intensive

in steady state but not unbounded. The financial-engineering question of how to size and finance the bond pool is downstream of the architecture proposal here.

Network effects and adoption. The architecture’s competitive position depends on adoption by a sufficient fraction of sub-suppliers and by passengers willing to trade the contract-of-carriage liability cap for the proportional bonded architecture. We do not model the adoption dynamics. The architecture is composable with the conventional regime: an operational party can serve some passengers under the bonded architecture and others under the contract-of-carriage regime, and a sub-supplier can serve some counterparties under bonded contracts and others under conventional ones. The transition is not all-or-nothing.

8 Conclusion

Scheduled passenger air service is a coordination problem across resource markets that the conventional architecture internalizes inside firm-shaped airlines because the transaction and enforcement costs of direct cross-market coordination were prohibitive. The bonded settlement primitive collapses those costs, and under the primitive the resource markets become directly accessible to the passenger through bonded commerce. The substantive contribution is not a better airline; it is a different way of organizing scheduled air service in which the firm-shaped container is structurally unnecessary because the resource markets it internalizes are directly coordinatable through bonded commerce. Cascading-delay reduction follows as a consequence of bond posture and atomic resolution; it is a property the architecture produces, not the property the paper is for.

The architecture is composable with existing safety regulation, crew-labor protections, insurance markets, and competition regimes. It changes the cost-flow path without displacing the regulatory and contractual layers that surround it. Whether the architecture diffuses into commercial aviation is an adoption question we do not adjudicate. What we do claim is that the architecture exists, that it is well-defined, and that its operational properties differ from the conventional regime in ways the operations-research literature on airline scheduling has been gesturing at without being positioned to deliver. The companion mechanism paper supplies the equilibrium analysis; the companion implementation paper supplies the verified Solidity implementation; the present paper supplies the assembly that turns those into a working scheduled-service coordination architecture.

A natural next step is empirical: a pilot implementation of the architecture for a domestic route with a manageable resource-market portfolio, with measurement of bond-loss frequency, disruption-resolution latency, and operational-improvement outcomes at each resource provider, compared to the conventional regime. The pilot would also surface the operational tooling questions inherent to a star-shaped process the kernel enforces: how passengers discover and select resource providers across the seven (or more) markets the journey draws on, what off-chain discovery / curation surfaces are useful, how cohort-internal compensation negotiations play out at IROps scale, and how multi-process composition operates when a resource provider itself bonds with sub-suppliers under their own `rootBuyer` role. We do not undertake the pilot here; we identify it as the experimental work the architecture invites.

Acknowledgments

The worked example originated in a project repository of agent-coordination scenarios; the present paper formalizes the scenario into an industrial-engineering register.

References

- Barnhart, C. and Cohn, A. Airline Schedule Planning: Accomplishments and Opportunities. *Manufacturing & Service Operations Management*, 6(1):3–22, 2004.
- Ball, M. O., Barnhart, C., Nemhauser, G., and Odoni, A. Air Transportation: Irregular Operations and Control. In C. Barnhart and G. Laporte, editors, *Handbooks in Operations Research and Management Science: Transportation*, volume 14, pages 1–67. Elsevier, 2007.
- Lan, S., Clarke, J.-P., and Barnhart, C. Planning for Robust Airline Operations: Optimizing Aircraft Routings and Flight Departure Times to Minimize Passenger Disruptions. *Transportation Science*, 40(1):15–28, 2006.
- Williamson, O. E. *The Economic Institutions of Capitalism*. Free Press, New York, 1985.
- European Parliament and Council. *Regulation (EC) No 261/2004 establishing common rules on compensation and assistance to passengers in the event of denied boarding and of cancellation or long delay of flights*. February 11, 2004.
- US Federal Aviation Administration. *14 CFR Part 117: Flight and Duty Limitations and Rest Requirements: Flightcrew Members*. 2014.